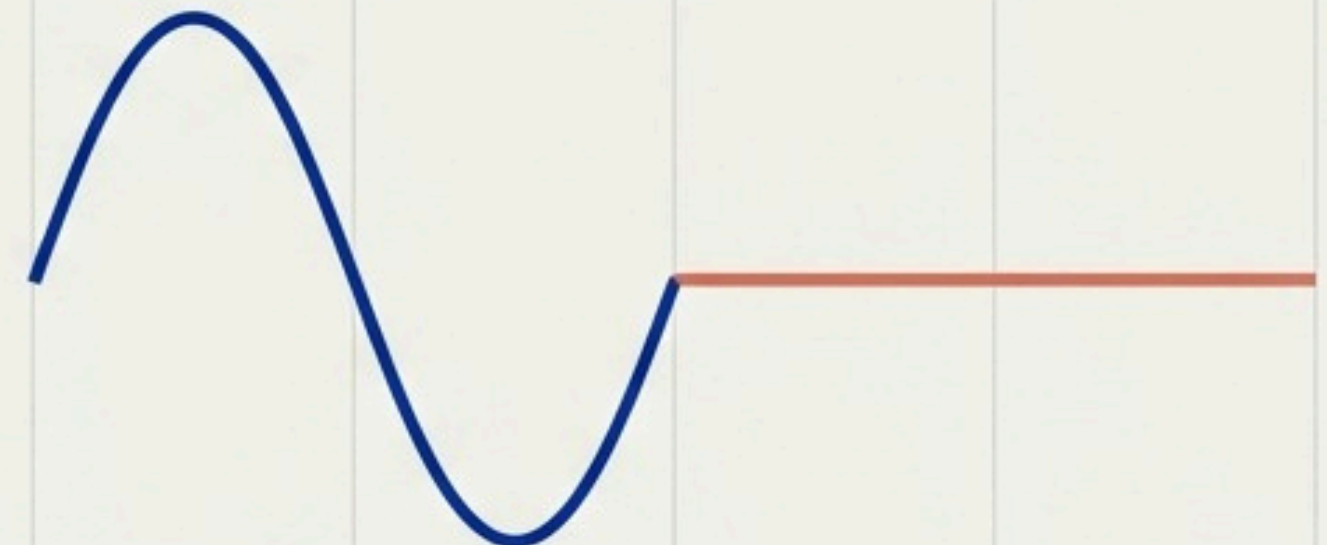


Test-Time Training (TTT)

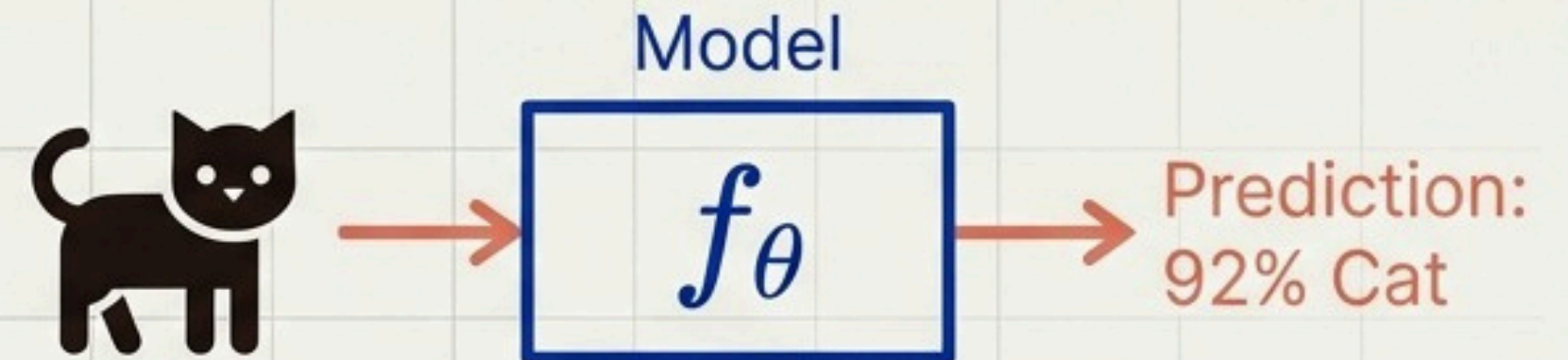
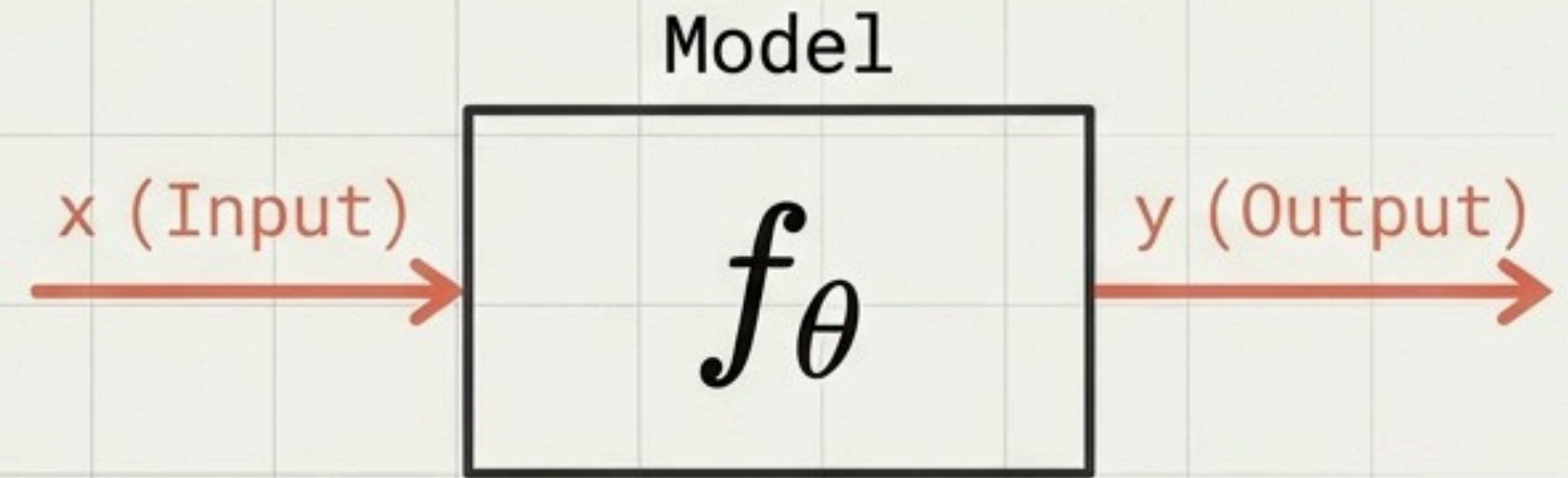
Adapting Models at Inference

Mehdi AGHAEI and Rayane KHATIM



What is a Model?

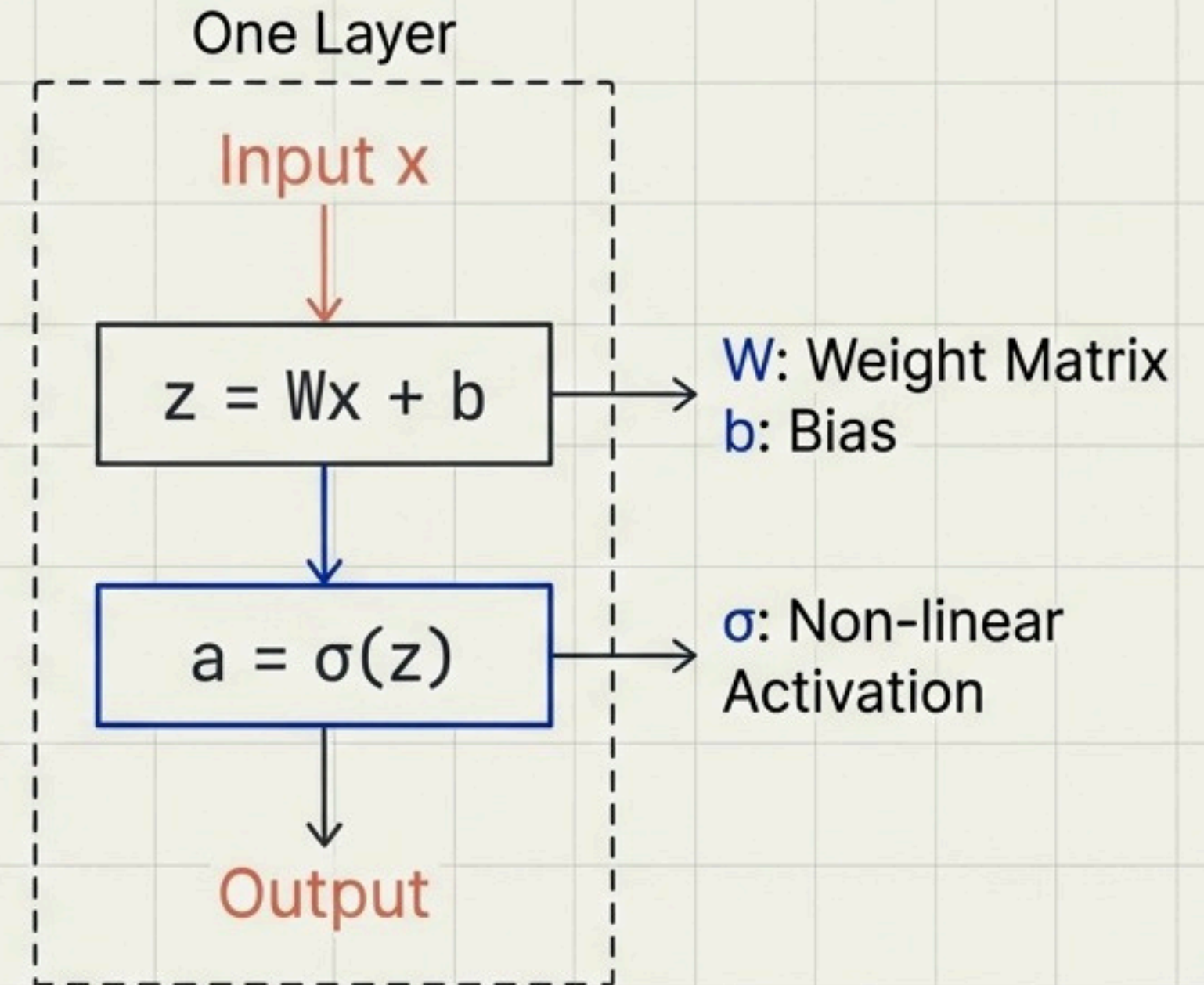
- A mathematical function mapping inputs to outputs.
- **Equation: $f_{\theta}(x) = y$**
- **x :** Input (e.g., image)
- **y :** Output (e.g., predicted class)
- **θ :** Parameters (adjustable coefficients)



The image (x) does not change.
The parameters (θ) change.

What is a Neural Network?

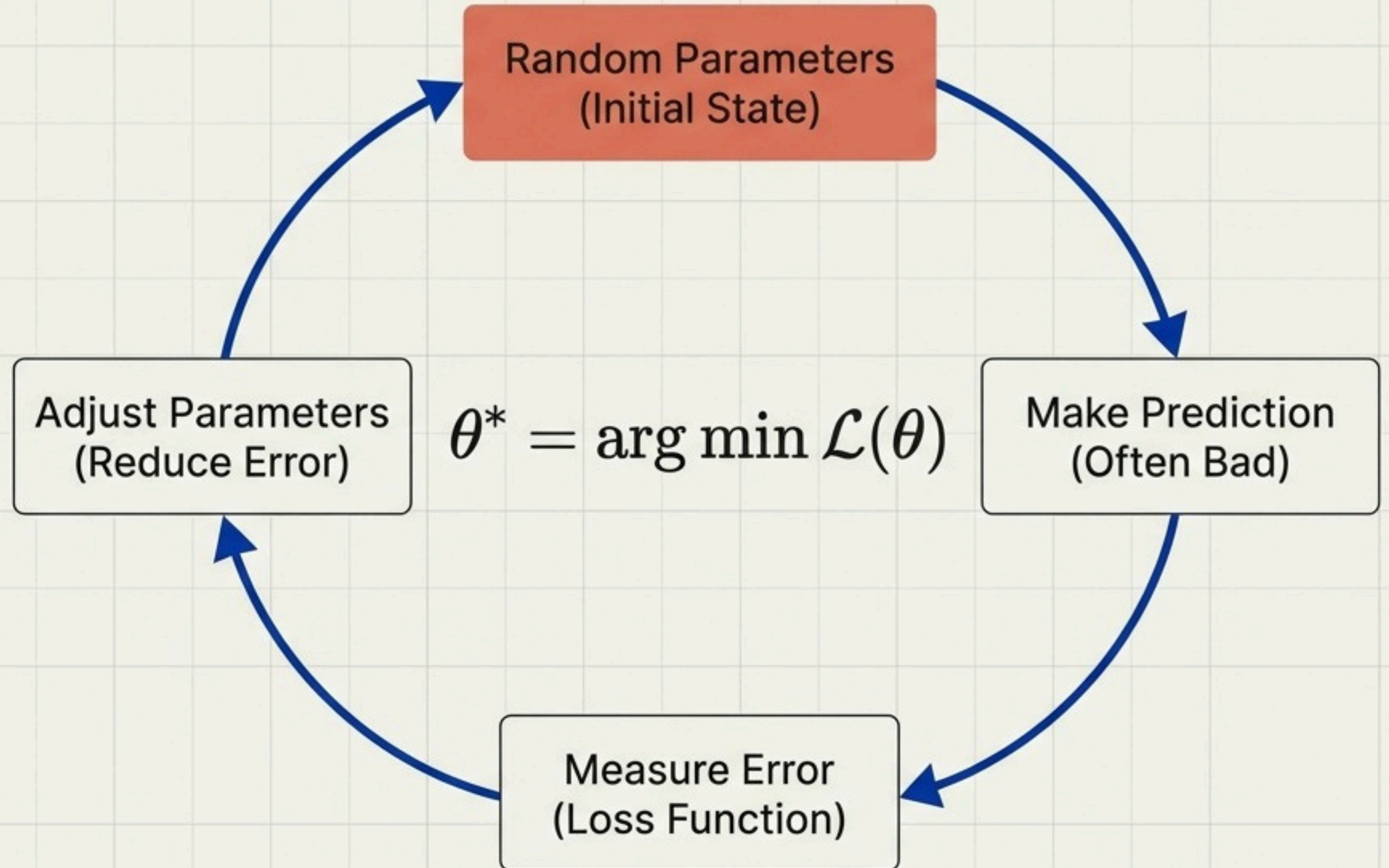
- Composition of multiple layers of parameterized functions.
- Equation: $f_{\theta}(x) = f_L(\dots f_1(x))$
- Layers perform three steps:
 - 1. Multiply inputs by weights (W)
 - 2. Add biases (b)
 - 3. Apply non-linear transformation (σ)



Important: The model does not “know” edges. Patterns emerge from training.

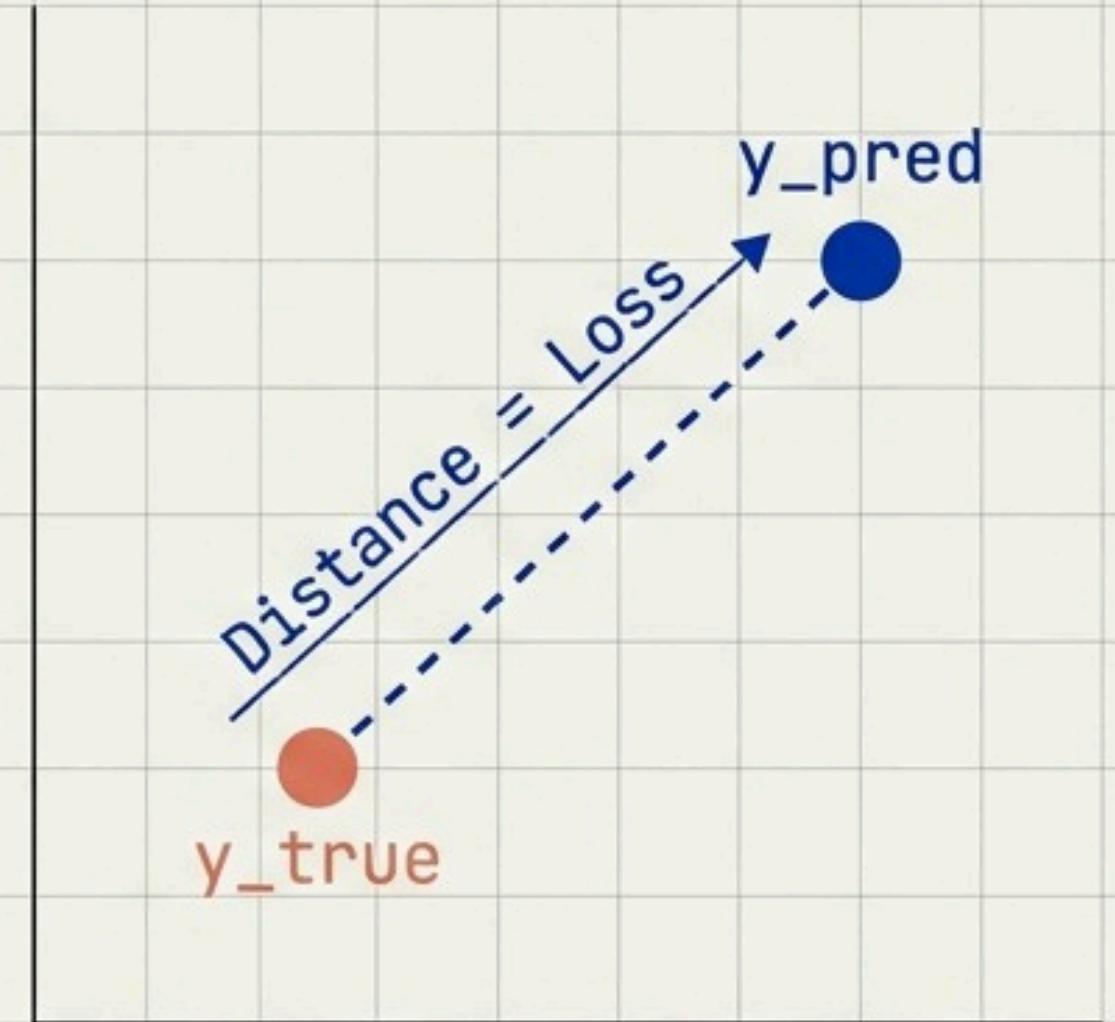
Training The Model

Training is the process of adjusting θ to minimize a loss function over a dataset.



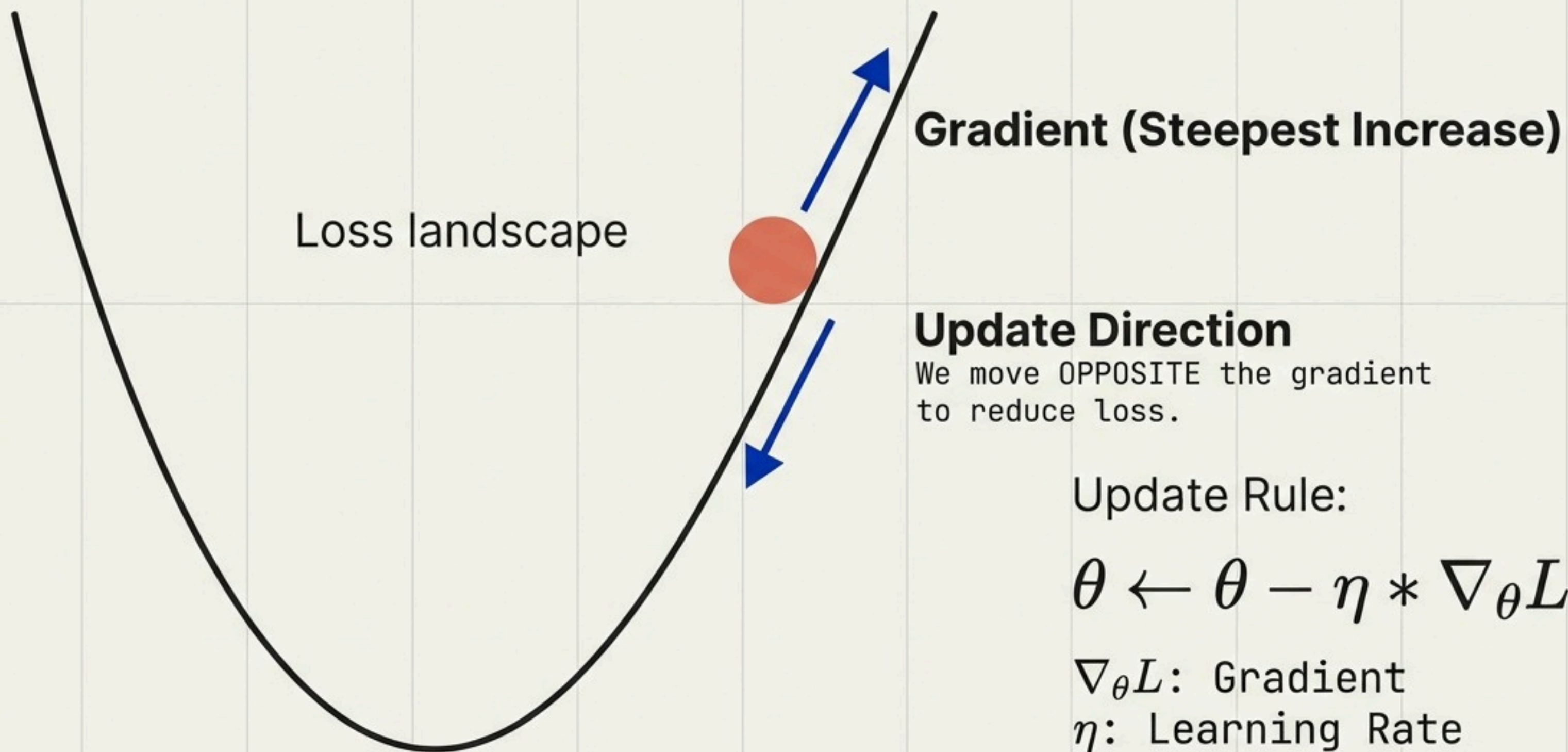
Loss Functions

- Loss is a function measuring the discrepancy between prediction and reality.
- Equation: $L(y_{\text{pred}}, y_{\text{true}})$
- Output: A real number indicating 'badness' (lower is better).
- Example: Euclidean distance between predicted and true values.



$$\text{Empirical Risk} = \frac{1}{m} \sum_{i=1}^m L(y_i, h(x_i))$$

Optimization: Gradient Descent



Standard Inference

Process of using a trained model to make predictions on new inputs.

The Normal Inference Rule:

- Parameters are FROZEN.
- No gradient computation.
- No weight updates.
- Forward pass only.



The model relies entirely on knowledge learned during training.

The Problem with Standard Inference

- Standard inference assumes “fixed behavior”.
- **Assumption:** Training data covers all future scenarios.
- **Reality:** Conceptual Shift. Real-world data distributions change.
- **Limitation:** Since parameters are frozen, no optimization happens on new data.

**Test-Time Training
(TTT) Idea:**

**Break the normal
inference rule. Allow
learning to happen AT
inference.**

The Solution: Self-Supervised Learning

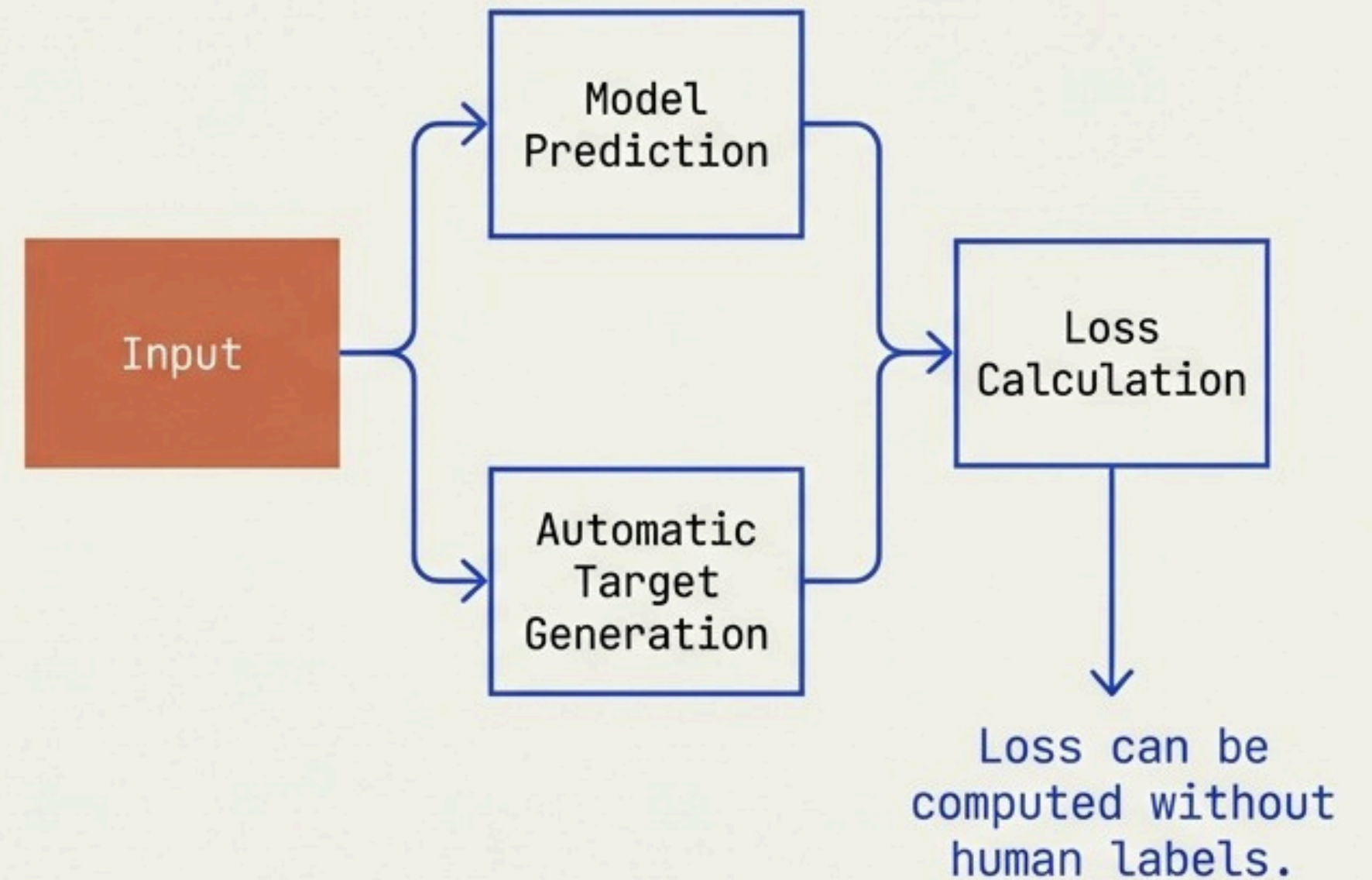
Learning Without Human Labels

Challenge: At inference, we have no human labels (y_{true}) to compute loss.

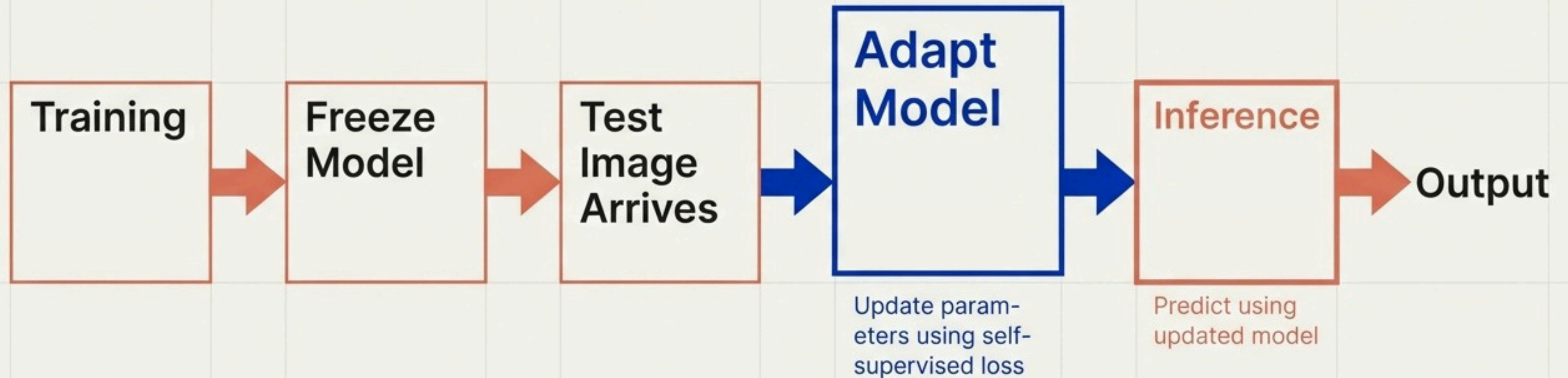
Solution: Self-Supervised Learning.

Mechanism: A target signal is generated AUTOMATICALLY from the input data.

Result: We can calculate a gradient and update parameters on the test image.

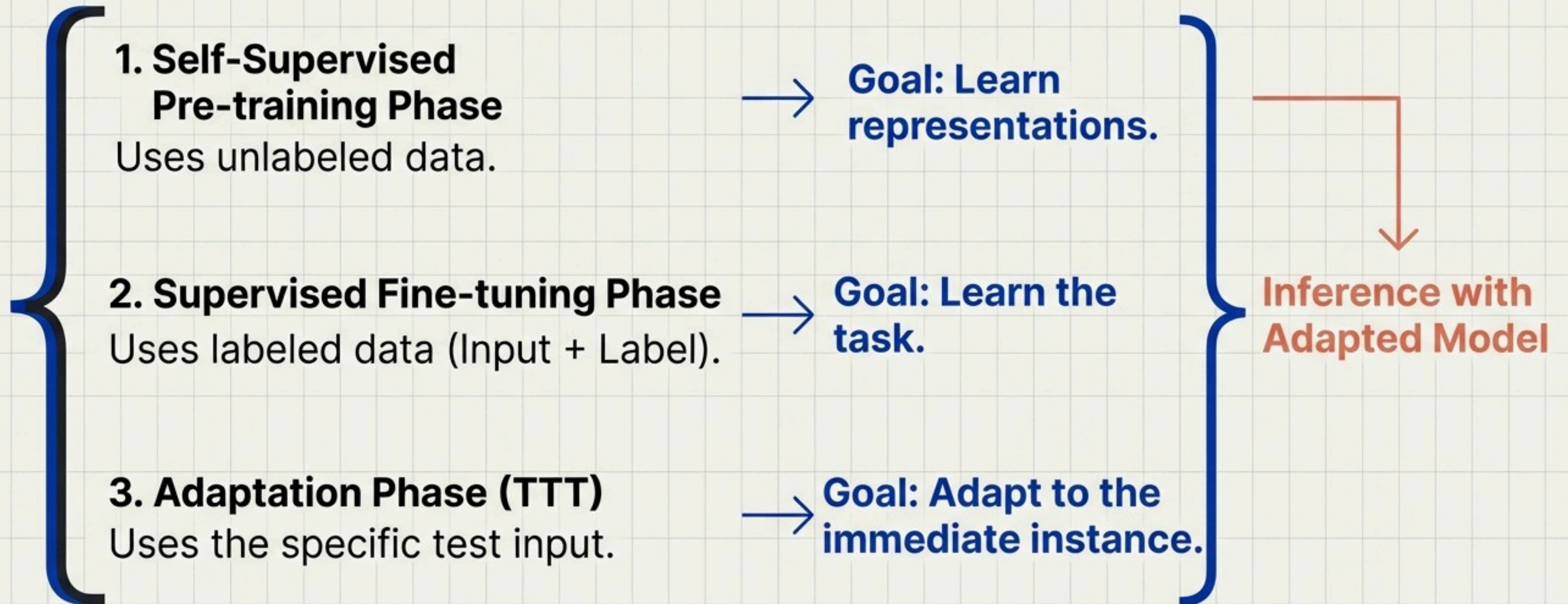


The TTT Pipeline



Normal Inference:	Input → Frozen Model → Output
TTT Inference:	Input → Adapt Model → Updated Model → Output

Three Distinct Datasets



Summary & Questions

Key Takeaways

- Standard inference uses frozen parameters.
- TTT updates parameters during test time.
- Self-supervised learning enables updates without labels.

Questions for Discussion

- If we update parameters for every test image, is it computationally expensive?
- Does the model 'forget' its original training after adapting?
- How exactly is the target signal constructed automatically?